

Necessary but Not Transplantable: The Multiplicative Gate of a Mamba-3 Block under Matched-Noise Cross-Condition Drift

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Abstract

The robustness of Mamba-style selective state-space models (SSMs) is often attributed to input-dependent selectivity. We test this attribution with a three-step ablation chain on one controlled testbed: a Mamba-3 block, bearing vibration signals (XJTU-SY, one speed-and-load transition), and noise-aware training at matched signal-to-noise ratio. Decision thresholds for each step were written down before the corresponding grid was run (internal pre-specification, not an external registry). First, replacing the input-dependent parts of Δ , A , B and C by learned constants—while the input-dependent trapezoidal weights and rotation angles of the Mamba-3 scan, the convolutional stem and the output gate are kept—costs at most about 5 pp under deep noise (paired intervals include zero), and the frozen block still outperforms a parameter-matched time-invariant S4D by 14–20 pp. Under the pre-specified criterion these four projections are therefore not the primary contributor. A second grid then freezes the trapezoidal weight and the rotation angles as well, leaving no tensor entering the scan dependent on the input: the advantage over S4D does not shrink but measures +21.9, +24.0 and +21.2 pp at 0, -2 and -6 dB (5/5 seeds, intervals excluding zero), with fewer parameters, so the block’s deep-noise advantage over a time-invariant SSM requires no input-dependent scan term at all. Second, removing the multiplicative SiLU gate from the frozen block costs 9–24 pp and returns it to roughly the level of S4D. Third, adding the gate to S4D recovers part of the gap, but not because of the gate: a narrow graft recovers 17–20%, a graft

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rebuilt in the donor’s width and projection structure recovers 30% (seed interval 16–52%; the pre-specified verdict is inconclusive between insufficient and partial recovery), and an additive branch with identical parameters recovers at least as much (41%). In the S4D host the multiplicative gate thus adds nothing beyond an equal-capacity additive branch, while its removal from the Mamba-3 block is costly; in this sense the gate’s benefit depends on the host. Conversely, replacing the gate inside the Mamba-3 block by an additive branch with identical parameters and initial weights rescues none of the loss (rescue ratio 0.02, seed interval upper bound 0.49; pre-specified verdict: not rescued): in its native host the multiplicative operator itself is required. On the reverse transition (the same bearings with their roles swapped) the fully input-independent scan again exceeds S4D by 22–26 pp and gate removal again costs 18–24 pp, all intervals excluding zero. A separate, shorter per-class grid shows the gate-related differences concentrating on the minority inner-race class. Values are best-epoch macro-F1 on the evaluation condition; the central contrasts for four arms are re-assessed under last-epoch reporting. The evidence is conditional on one block, one pair of operating conditions run in both directions with the same eight bearings, five seeds and matched-noise training.

Keywords: state-space models, Mamba, input selectivity, gating, robustness, ablation, distribution shift, matched-noise training

1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet it sits in unresolved tension with recent evidence: the first systematic S4D-versus-Mamba comparison on time-series classification finds the *simpler, time-invariant* model consistently superior [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain

15 whose decision thresholds were fixed before each grid was run. The testbed
 16 is bearing vibration diagnosis under operating-condition drift and matched
 17 noise, in which the robustness gap is large (up to 18 pp), the evaluation la-
 18 bels are identical across all 865 cells (hash-verified), and the block can be
 19 dissected.

20 Our approach is deliberately *attributional* rather than *methodological*: we
 21 propose no new architecture and claim no state-of-the-art. Instead we take
 22 a regime in which a Mamba-style block exhibits a large, stable robustness
 23 advantage over a time-invariant SSM, and ask which ingredient earns it.
 24 The chain has three steps, each with adjudication thresholds fixed before the
 25 corresponding grid was run:

- 26 **1. Test the classical selectivity terms (necessity test).** We con-
 27 struct a *frozen* arm: the input-dependent parts of Δ , A , B and C
 28 are replaced by learned constants. The Mamba-3 scan also takes an
 29 input-dependent trapezoidal weight and input-dependent rotation an-
 30 gles; these, the convolutional stem, the multiplicative SiLU gate, the s-
 31 tate parameterisation, normalisation and projections are retained (Sec-
 32 tion 3.2). If these four terms drive robustness, the frozen arm should
 33 collapse toward the time-invariant baseline.
- 34 **2. Localise the necessary component.** On the frozen base, we remove
 35 one component at a time (convolutional stem; SiLU gate; heavy-tailed
 36 state parameterisation) under a single-variable discipline checked key-
 37 by-key on the state dictionary (Section 3.2), and adjudicate each arm
 38 against pre-specified Δ -thresholds.
- 39 **3. Test sufficiency by constructive grafting.** The component iden-
 40 tified as necessary is added to plain S4D as a gate branch, and a pre-
 41 specified recovery ratio decides whether “removal collapses” is matched
 42 by “addition restores”.

43 The outcome is a conditional pattern. Freezing the input-dependent Δ ,
 44 A , B , C projections costs at most about 5 pp in deep noise (paired intervals
 45 include zero) while the block retains a 14–20 pp advantage over S4D (best-
 46 epoch and last-epoch reporting, Section 4.8): under the pre-specified criterion
 47 these projections are not the primary contributor. Freezing the trapezoidal
 48 weight and the rotation angles as well—so that nothing entering the scan
 49 depends on the input—does not shrink the advantage either: it measures
 50 +21 to +24 pp over S4D with fewer parameters (Section 4.2). Removing the

SiLU gate costs 9–24 pp and takes the block to roughly the level of plain S4D or below. Adding a gate branch to S4D recovers only $R = 0.17$ of the gap (0.20 under last-epoch reporting). Rebuilt in the donor’s width and projection structure, the graft recovers $R = 0.30$ (interval 0.16–0.52), but an additive branch with identical parameters recovers at least as much (0.41, Section 4.5). In the S4D host, then, the multiplicative gate adds nothing beyond an equal-capacity additive branch, whereas in the Mamba-3 block its removal is costly; we read this as a host dependence of the gate’s benefit in this design. The converse test closes the argument: replacing the gate inside the Mamba-3 block by an additive branch with identical parameters and initial weights rescues none of the loss (rescue ratio 0.02, Section 4.6). The gate is necessary as a multiplicative operator in its native host and is not transplanted with it. The first two steps replicate on the reverse transition (Section 4.7). A per-class analysis on a separate, shorter grid indicates that the gate-related differences concentrate on the minority class.

Contributions. (i) A layered freezing experiment on a Mamba-3 block under distribution drift and matched noise, from the classical selectivity terms (Δ , A , B , C) through to a scan in which no tensor depends on the input, in which the block’s deep-noise advantage over S4D survives every step; each arm is verified input-independent by capturing the tensors the scan receives for two different inputs. (ii) Component-level evidence that the output gate is the one ablated component whose removal causes a large, one-signed loss, and that adding it to a time-invariant host, even in the donor’s width and with an additive control of identical size, recovers no more than the added structure does, while replacing it in its own block by an additive branch of identical size recovers nothing; both of the first two findings replicate on the reverse transition. (iii) A reporting-rule sensitivity analysis (best epoch on the evaluation condition versus final epoch) showing that absolute levels shift by 1.4–9.5 pp while the central contrasts keep their sign. (iv) A documented pre-specification and review trail—evaluation-label hashes, thresholds written before each grid, and agent-assisted review—including the incidents in which automated adjudication departed from the pre-specified criteria; we release the trail.

2. Background and Related Work

2.1. Time-invariant and selective SSMs

Diagonal structured SSMs (S4D) [6] evolve a linear state with input-independent parameters (A , B , C , Δ), acting as a bank of learned linear filters. Mamba-style blocks [1] make Δ , B , C functions of the input token (selectivity), wrap the scan in a block containing a short causal convolutional stem, a multiplicative SiLU gate on a parallel branch, and residual/normalisation structure; the concrete block used throughout this paper follows the Mamba-3 formulation [7]. The narrative crediting selectivity for robustness draws on theory showing input-dependent dynamics can represent discontinuity-sensitive projections and counteract memory decay [8], and on [9]. We treat these as *hypotheses to be tested*, not established attributions: the theoretical results concern representability, which does not by itself determine which component earns empirical robustness in a trained block.

2.2. The unresolved comparison

Empirically the picture is contested. A systematic 59-dataset comparison finds S4D-based models superior to Mamba variants on clean time-series classification [4]; theoretical work identifies long-range regimes where Mamba is weaker than S4D [5]. Meanwhile, application literatures—including machine-condition monitoring—contain many Mamba-hybrid methods reporting noise robustness [10, 11, 12, 13, 14], but none, to our knowledge, isolates *which component* of the block is responsible under controlled conditions. Our clean-data results agree with the S4D-superiority finding (S4D matches or slightly exceeds the Mamba-style block on clean data). Whether the disagreement in the literature reflects a regime difference is not settled by a single testbed; what we can report is that, in the drift $\times$ noise regime studied here, the Mamba-style block’s advantage over S4D is not removed by freezing the classical selectivity terms.

2.3. Testbed rationale

We use bearing vibration diagnosis under cross-condition drift (speed and load both change) as the testbed, for three properties that matter to the attribution question rather than to the application: (a) the distribution shift is physically generated (a change of rotation speed and radial load between recorded conditions), not simulated; (b) fault signatures are sparse transients—precisely the discontinuous structures selectivity theory predicts

selective SSMS should excel at [8], which makes it a plausible arena for the selectivity hypothesis; and (c) the robustness gap between the Mamba-style block and S4D is large (up to 20 pp), giving the attribution chain high signal. The testbed is two-class (details in Section 3.1); the scope of all claims is bounded accordingly (Section 6).

3. Experimental Protocol

3.1. Testbed, arms, and budget

Vibration windows from the XJTU-SY run-to-failure dataset [15] are split by operating condition: models train on one operating condition (37.5 Hz, 11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change (cross-condition), with additive white Gaussian noise applied at matched SNR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The task is binary fault classification (inner-race vs. outer-race dominant classes; class imbalance makes IR the hard class, Section 4.9). All arms share an identical training budget (50 epochs, 5 seeds; the per-class grid of Section 4.9 used 20 epochs), identical splits, and identical evaluation labels, verified by a hash of the label vector: every reported cell carries `eval_sha = 6c20b367...` (Appendix A), recomputed independently during adjudication. The hash certifies the label vector, not the input windows. In total the study comprises 865 cells across five pre-specified grids (baseline grid 475; frozen-selectivity 105; component ablation 175; graft 35; per-class 75), and four follow-up grids add 255 cells (Sections 4.2, 4.8, 4.5, 4.6 and 4.7).

Composition of the data. Training uses the bearings of the 37.5 Hz/11 kN condition whose failure mode is inner race or outer race (Bearing2_1 is labelled IR; Bearing2_2, 2_4 and 2_5 are labelled OR); evaluation uses those of the 40 Hz/10 kN condition (Bearing3_3 and 3_4 are IR; Bearing3_1 and 3_5 are OR). Bearings with cage or mixed failure modes are excluded. Windows are non-overlapping segments of 2048 samples of the horizontal channel taken from the fault phase, which is delimited by a kurtosis/RMS onset criterion with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR windows, the training set 4,624 OR and 624 IR windows. Two consequences matter for interpretation. The IR class is learned from a single training bearing, so class-level results may reflect bearing-specific generalisation. And the five seeds resample initialisation, batch order and noise but not bearings: there are four independent bearings on each side, and none of the intervals

154 reported here captures bearing-level uncertainty. One follow-up grid (Sec-
 155 tion 4.7) reverses the transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). It uses the
 156 same eight bearings with their roles swapped, so it tests the other direction
 157 rather than independent bearings; its evaluation set holds 4,624 OR and
 158 624 IR windows, all IR windows coming from Bearing2_1, and its label hash
 159 is 93424834.... The 35 Hz/12 kN condition has no inner-race bearing and
 160 cannot form a binary pair.

161 The principal arms are summarised in Table 1.

162 3.2. What the frozen arm freezes, and what the checks verify

163 In the Mamba-3 layer used here, one input projection emits the gate and
 164 value streams, the scan inputs B and C , the step-size and decay terms (Δ
 165 and A , through input-dependent additive terms), a trapezoidal weight and
 166 rotation angles. The frozen arm removes the input-dependent parts of Δ ,
 167 A , B and C and replaces them with learned constants broadcast over time
 168 (for Δ the static bias of the original layer is kept; for A a new constant pa-
 169 rameter plays the role of a static log-decay, passed through the layer’s own
 170 heavy-tailed clamp). The removed projections are deleted, so no parameter
 171 or gradient can leak. **The trapezoidal weight and the rotation angles**
 172 **remain functions of the current input**, as do the stem and the output
 173 gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3
 174 formulation [7] the input-dependent rotation and trapezoidal weighting also
 175 enter the state update. Section 4.1 is accordingly a statement about the
 176 classical selectivity terms Δ , A , B , C only. Three further arms close that gap
 177 by the same device: *frozen-trap* replaces the trapezoidal weight by a learned
 178 per-head constant, *frozen-rot* replaces the rotation angles by learned con-
 179 stants (so the rotation becomes a learned position-only phase rather than a
 180 function of the current input), and *frozen-all* does both, leaving **no tensor**
 181 **entering the scan dependent on the input**. In each case the correspond-
 182 ing `in_proj` output slice is deleted. Section 4.2 reports them. For these arms
 183 the construction check adds the decisive probe: the tensors handed to the
 184 scan are captured for two different inputs through the same module, and a
 185 tensor the arm claims to have frozen is required to be bitwise identical across
 186 them while a tensor left selective is required to differ (all three arms pass).
 187 No kernel modification is needed, because the scan interface accepts these
 188 tensors regardless of their origin.

189 Single-variable discipline was checked by key-level state-dictionary diffs
 190 between two models built under the same seed. For the frozen arm the

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ , A , B , C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Section 4.3)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Section 4.3)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Section 4.3)	112,410
frozen-trap	Frozen base with the trapezoidal weight also made a learned constant (Section 3.2)	111,906
frozen-rot	Frozen base with the rotation angles also made learned constants	104,346
frozen-all	Frozen base with both also constant: no tensor entering the scan depends on the input	103,842
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Section 4.4)	116,802
S4D+gate (wm)	Graft rebuilt in the donor’s shape: per layer x, z projected to width 128, S4D at width 128 (state 64, same SSM budget as the host), $W_{\text{out}}(y \odot \text{SiLU}(z))$ (Section 4.5)	198,978
S4D+branch	Same modules, parameters and initial weights, combined additively: $W_{\text{out}}(y + \text{SiLU}(z))$ (control)	198,978
frozen+add	BM3-frozen with the gate replaced in place: $W_{\text{out}}(y + \text{SiLU}(z))$ instead of $W_{\text{out}}(y \odot \text{SiLU}(z))$; same modules ⁸ parameters and initial weights (Section 4.6)	112,410
CNN/TCN baselines	Conventional baselines for context	—

diff verifies that every non-target tensor has the same name and shape as in BM3; for the component-ablation and graft arms it additionally compares the values of non-target tensors at initialisation. Every differing key is mapped to an entry in an architecture map. We do not claim byte-level identity of non-target tensors in the frozen arm, and trained weights naturally differ between arms; the checks concern how the arms were constructed.

3.3. Reporting rule

The macro-F1 reported for each cell is the *maximum over epochs* of macro-F1 evaluated on the evaluation split, i.e., the held-out operating condition that every arm is scored on (the split covered by the fingerprint above). The harness defines no separate validation split, so the epoch is chosen with the evaluation data. Two consequences follow. First, absolute values are optimistically biased, by an amount that grows with an arm’s epoch-to-epoch variability. Second, a common rule cancels a shared offset between arms but not an arm-dependent one. The pre-specified thresholds were fixed against this rule, so all adjudications in Section 4 are reported under it, unchanged. Section 4.8 then re-assesses the central contrasts under a rule that does not use the evaluation split (the final epoch), and the boundaries section states what that re-assessment does and does not cover.

3.4. Confound elimination

Before component attribution, three candidate confounds were examined on the same grid. **Capacity:** S4D-wide ($1.75\times$ parameters) tracks S4D pointwise (mean -0.5 pp), so this particular increase in S4D’s parameter count does not explain the gap; S4D-wide was not re-run under last-epoch reporting, and the control does not address the change in channel mixing that gate removal also entails. **Normalisation:** both SSM arms use LayerNorm and contain no BatchNorm; the arm that degrades under noise is the one *without* any batch statistics, which argues against BatchNorm statistics mismatch as the explanation. Other normalisation inside the Mamba-3 layer was not varied. **Auxiliary physics loss:** an exploratory kinematics-alignment regulariser carried by the original model contributes -0.1 pp pointwise (curves coincide); all attribution arms are therefore built on the pure cross-entropy variant.

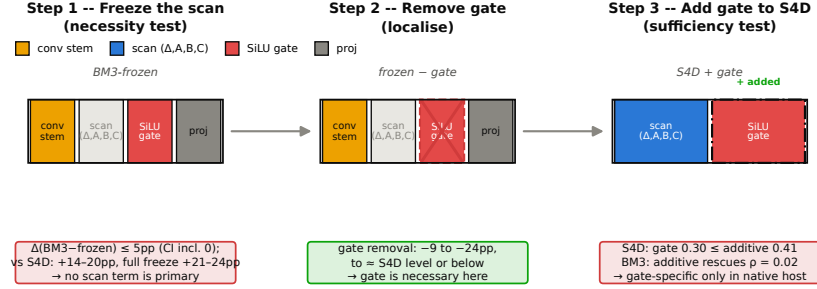
224 3.5. *Pre-specification and adjudication*

225 Every adjudicated question in this study had numeric decision thresholds
226 written to a task card and to disk before the corresponding grid was launched.
227 This is internal pre-specification, not external registration: the supporting
228 evidence is local file modification times and run-directory names (released,
229 [Appendix A](#)), which are not tamper-evident and are not preserved by a
230 repository checkout, so the trail lets readers inspect the order of events but
231 cannot prove it. The exception is the last two follow-up grids (Sections 4.5–
232 4.7): their thresholds, and for the last grid the title decision that depends on
233 its outcome, were committed to the public repository before the first cell was
234 run, which gives those grids an external time-stamp. Later questions (com-
235 ponent ablation, graft) were specified after earlier grids had been seen, as is
236 natural in a staged design. Adjudication was mechanical: an analysis pass
237 applies the thresholds and emits a decision artifact; an independent reviewer
238 pass re-computes aggregates, verifies fingerprints row-by-row, and must cite
239 file paths and modification times for every check. The pre-specified thresh-
240 olds were binding in practice: two attempted post-hoc criterion substitutions
241 and one stale-artifact false verification were caught and re-adjudicated under
242 the original rules — this is the one place where pre-specification actually
243 bit. [Appendix C](#) details the incidents, the automated executor/reviewer
244 implementation, and the human adjudication gates; all scientific decision-
245 s — thresholds, rulings, and interpretations — were made by the authors
246 and are timestamped. Statistical tests (paired, per-condition) are report-
247 ed throughout but were never adjudication criteria. A single repeat of one
248 2-epoch configuration in a separate process differed by ~ 0.7 pp macro-F1
249 (non-deterministic kernel reductions; [Appendix B](#)); this is an observed re-
250 peat difference, not an estimated variability bound, and arm-level contrasts
251 are judged against seed-level variability (2–9 pp).

252 4. Results

253 Throughout, “deep noise” denotes the pre-specified adjudication band
254 $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported but excluded from adjudication (the time-
255 invariant baseline is near chance there: a constant majority-class predictor
256 scores 40.0% and a uniformly random predictor 48.6% macro-F1 on this
257 evaluation set, against 52.3% for S4D). All values are macro-F1 (%), mean
258 over 5 seeds, best-epoch on the evaluation condition (Section 3.3) unless

259 stated otherwise. Figure 1 summarises the three-step ablation chain that
 260 structures this section.



Conditional on this block, testbed and budget: the gate's benefit depends on the host.

Figure 1: The three-step ablation chain with pre-specified thresholds: freeze the input-dependent Δ, A, B, C projections (Section 4.1), remove one component at a time on the frozen base to localise the necessary ingredient (Section 4.3), and graft that ingredient onto S4D to test sufficiency (Section 4.4).

261 4.1. Freezing the input-dependent Δ, A, B, C projections

Table 2: BM3 vs. BM3-frozen vs. S4D, macro-F1 (%) by SNR (frozen-selectivity grid). $\Delta(\text{BM3}-\text{frozen})$ and $\Delta(\text{frozen}-\text{S4D})$ in percentage points.

Condition	BM3	BM3-frozen	S4D	$\Delta(\text{BM3}-\text{frozen})$	$\Delta(\text{frozen}-\text{S4D})$
clean	78.9	78.9	81.0	-0.0	-2.1
0 dB	95.0	89.9	76.1	+5.1	+ 13.8
-2 dB	95.4	92.1	75.1	+3.3	+ 17.0
-6 dB	88.5	86.3	68.5	+2.2	+ 17.7
-10 dB*	68.1	64.3	52.3	+3.8	+12.0

262 Freezing the input-dependent Δ, A, B, C projections (Section 3.2) costs
 263 at most 5.1pp anywhere in the adjudication band on the point estimates,
 264 while the frozen block retains a 13.8-17.7pp advantage over S4D (Table 2,
 265 Figure 2). Five seeds bound these quantities loosely: paired 95% intervals
 266 for $\Delta(\text{BM3}-\text{frozen})$ are $[-4.7, 15.0]$, $[-3.1, 9.8]$ and $[-7.5, 11.8]$ pp at 0, -2
 267 and -6 dB (BM3 ahead in 3/5, 4/5 and 2/5 seeds), and for $\Delta(\text{frozen}-\text{S4D})$

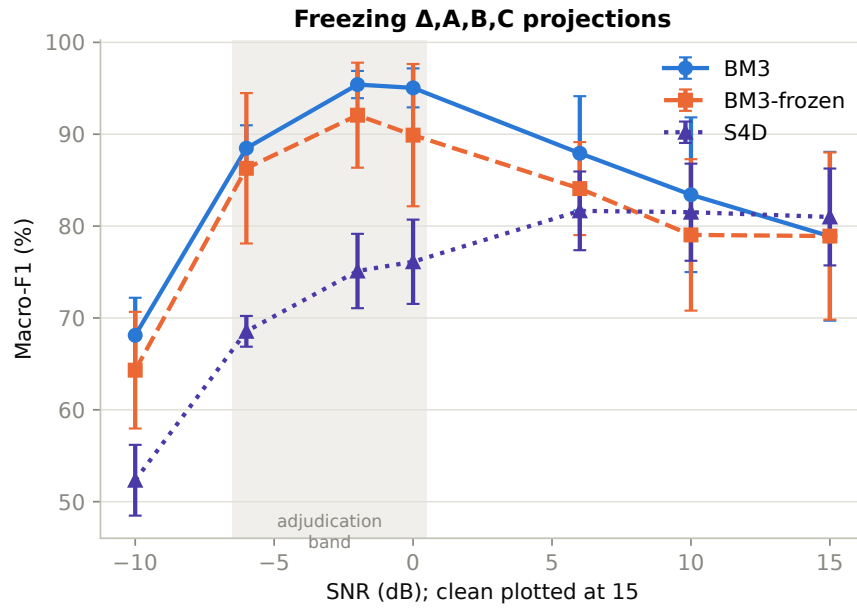


Figure 2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid. The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band.

268 $[-0.5, 28.0]$, $[5.5, 28.4]$ and $[5.8, 29.7]$ pp (5/5, 5/5, 5/5 seeds). The pre-
 269 specified rule attributed robustness to selectivity only if $\Delta(\text{BM3-frozen})$
 270 ≥ 10 pp in deep noise with the frozen arm collapsing toward S4D. Neither
 271 condition holds on the point estimates, so the pre-specified verdict is that
 272 these projections are *not the primary mechanism*. This is a failure to meet
 273 a confirmation criterion, not a demonstration that freezing is cheap: the
 274 intervals above are compatible with differences larger than 10 pp, and no e-
 275 quivalence margin was pre-specified or tested. One limit applies: the frozen
 276 arm still contains input-dependent trapezoidal weights and rotation angles,
 277 so this result concerns Δ, A, B, C only; Section 4.2 removes that limit. Un-
 278 der last-epoch reporting the re-run gives $\Delta(\text{BM3-frozen}) = +3.7, +0.9$ and
 279 $+0.2$ pp (best-epoch re-run: $+5.1, +3.5, +3.0$ pp; intervals still include zero
 280 at every level), so the point estimates fall further below the 10 pp threshold
 281 rather than rising towards it. On clean data S4D matches or slightly exceeds
 282 both Mamba-style arms—consistent with the systematic clean-TSC compar-
 283 ison [4]. In the noisy regime, the block’s advantage over S4D is not removed
 284 by freezing these four terms, so it must be carried by other parts of the block
 285 or by the remaining input-dependent scan terms.

286 4.2. Freezing the remaining input-dependent scan terms

287 Table 3 reports the three layered arms of Section 3.2 on the adjudication
 288 band (60 cells: 4 arms $\times$ 3 levels $\times$ 5 seeds $\times$ 50 epochs, the original training
 289 code, the same evaluation fingerprint, with the original BM3 arm re-run
 290 in the same grid). The question is whether the advantage that survived
 291 freezing Δ, A, B, C is carried by the two input-dependent terms the frozen
 292 arm retained.

293 It is not. With every tensor entering the scan input-independent, *frozen-all*
 294 is $+21.9, +24.0$ and $+21.2$ pp above S4D at 0, -2 and -6 dB under final-
 295 epoch reporting, positive in 5/5 seeds at every level and with intervals ex-
 296 cluding zero—if anything a larger margin than the partially frozen arm’s
 297 $+15.5, +19.8, +16.4$ pp (Table 10). None of the three freezing steps costs
 298 accuracy: the paired differences $\text{BM3-frozen} - \text{frozen-trap}$, $-\text{frozen-rot}$
 299 and $-\text{frozen-all}$ are negative at all three levels (between -1.8 and -6.3 pp
 300 under final-epoch reporting), i.e. the more heavily frozen arm is numerically
 301 ahead, with every interval including zero. The direction is not an artefact of
 302 capacity: *frozen-all* has 103,842 parameters against 112,410 for BM3-frozen
 303 and 177,938 for BM3. At the deepest level *frozen-all* also exceeds the full
 304 BM3 block by 4.5 pp ($[-7.2, -1.9]$ for BM3 $-\text{frozen-all}$, 0/5 seeds in BM3’s

Table 3: Layered freezing of the scan’s input-dependent terms. Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting; Δ columns are paired differences under final-epoch reporting with 95% t -intervals over seeds.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3 (re-run)	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen (Δ, A, B, C)	90.6	83.1	92.3	87.3	85.7	83.1
frozen-trap	95.5	87.6	95.3	89.3	90.5	88.0
frozen-rot	94.6	86.7	94.6	89.1	89.0	86.4
frozen-all	96.5	89.5	96.8	91.5	90.5	87.9
S4D	75.9	67.6	75.1	67.5	68.6	66.7
<i>Paired differences, final-epoch reporting (pp)</i>						
frozen-all − S4D	+21.9 [16.7, 27.1]		+24.0 [21.4, 26.5]		+21.2 [17.5, 24.9]	
BM3-frozen − frozen-all	−6.3 [−17.6, 4.9]		−4.1 [−12.0, 3.7]		−4.8 [−17.8, 8.2]	
BM3 − frozen-all	−2.6 [−8.2, 2.9]		−3.2 [−7.0, 0.6]		−4.5 [−7.2, −1.9]	

305 favour); we report this but do not build on it, since five seeds and one speed
306 pair cannot separate a genuine advantage of a time-invariant scan from a
307 variance effect.

308 Two readings of the whole step are therefore available, and the data do not
309 separate them: input-dependence of the scan contributes nothing measurable
310 here, or it contributes something that the learned constants recover under this
311 training budget. Either way the conclusion for the attribution chain is the
312 same and now rests on a complete intervention rather than a partial one: **the**
313 **Mamba-style block’s deep-noise advantage over a time-invariant**
314 **S4D does not require any input-dependent term in its scan.** What
315 remains to carry it is the rest of the block—the convolutional stem, the heavy-
316 tailed A parameterisation, the normalisations and the output gate—which
317 is what Sections 4.3–4.4 examine. The re-run also reproduces the original
318 grids: the BM3 arm’s best-epoch means (95.6, 95.8, 88.6) exceed Table 2’s
319 95.0, 95.4, 88.5 by 0.57, 0.43 and 0.16 pp (a different PyTorch/Triton build,
320 Section 4.8).

321 4.3. Removing the gate is the only ablation with a large, one-signed effect

322 “Necessary” is used here in the operational sense of this ablation: remov-
323 ing the gate from the frozen block, with the other components and the budget
324 unchanged, removes most of the block’s advantage. The gate result is large,
325 one-signed and monotone in noise depth: removal costs 13.1, 21.1 and 23.5 pp

Table 4: Single-component removal on the frozen base (component-ablation grid, trained separately from Table 2), $\Delta(\text{frozen} - \text{arm})$ in percentage points.

Component removed	0 dB	-2 dB	-6 dB	Adjudication
SiLU gate	+13.1	+21.1	+23.5	PRIMARY (all ≥ 8 pp)
conv stem ($k=7$)	-6.6	-2.7	+7.9	outside pre-specified bins (regime trade-off)
heavy-tailed $A \rightarrow$ S4D-style A	-2.4	-2.2	-4.2	outside bins (removal numerically beneficial; intervals include 0)

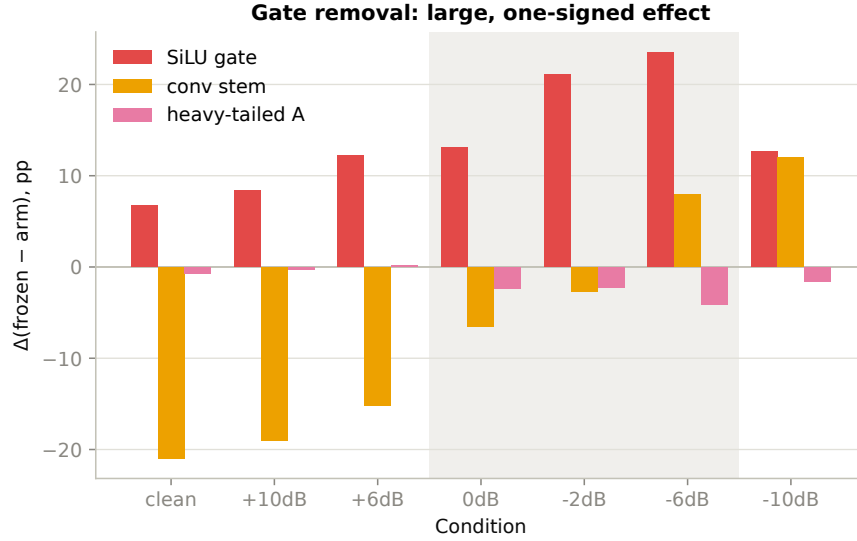


Figure 3: Component-removal Δ (percentage points, frozen - arm) by SNR for the three single-component ablations on the frozen base. Only gate removal is monotone, large, and one-signed across the adjudication band.

at 0, -2 and -6 dB (paired 95% intervals $[6.5, 19.6]$, $[3.8, 38.5]$, $[8.5, 38.6]$ pp; 5/5 seeds at every level). The gateless arm also gives up most of the Mamba advantage: it sits **below plain S4D** at -2 dB (71.7 vs. 75.2) and -6 dB (62.5 vs. 68.6), though not at 0 dB (77.6 vs. 76.2), and the gateless–S4D gap itself is not resolved by five seeds (Section 4.8: under last-epoch reporting the gateless arm is below S4D at -6 dB only). We read this as “at best on par with S4D, and worse at the deepest noise level”, which suggests, but does not establish, that components interact rather than add. The other two components tell regime stories rather than robustness stories: the convolutional stem helps by 7.9 pp at -6 dB (removal gives 78.1 vs. 86.0; paired 95% interval of the difference $[-2.1, 17.9]$ pp, 4/5 seeds) but *costs* 21.0 pp on clean cross-speed data (removal yields 98.8 vs. 77.8, with seed-std collapsing from 8.1 to 0.6 pp)—consistent either with speed-specific local temporal patterns mistransferring across conditions, or with optimisation friction under the matched budget; we report both readings. Replacing the heavy-tailed state parameterisation by an S4D-style one is numerically beneficial at all three deep-noise levels ($+2.4$, $+2.2$, $+4.2$ pp), but every paired interval includes zero (e.g., $[-15.4, 7.1]$ pp at -6 dB), so we draw no directional conclusion from it (numerically, the S4D-style variant is the best deep-noise cell in the component grid, 90.1 at -6 dB).

4.4. Adding a narrow gate branch to S4D

Table 5: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added). The S4D+gate column is the graft grid; the S4D and BM3-frozen columns, and hence R , are the component-ablation grid of Section 4.3, as pre-specified, which is why they differ from Table 2 by 0.1–0.7 pp (independent runs of the same arms).

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
0 dB	79.8	76.2	90.6	0.249
-2 dB	79.4	75.2	92.8	0.239
-6 dB	69.2	68.6	86.0	0.035

Pre-specified recovery ratio $R = \text{mean over the band of } (\text{S4D+gate} - \text{S4D}) / (\text{frozen} - \text{S4D}) = \mathbf{0.174}$. The pre-specified bins were $R \geq 0.7$ (confirmed), $0.3 \leq R < 0.7$ (partial recovery) and $R < 0.3$ (insufficient); $R = 0.174$ falls in the last. Resampling seeds within each arm gives a 95% interval of $[0.04, 0.29]$ for R , and the per-level paired gains of S4D+gate over

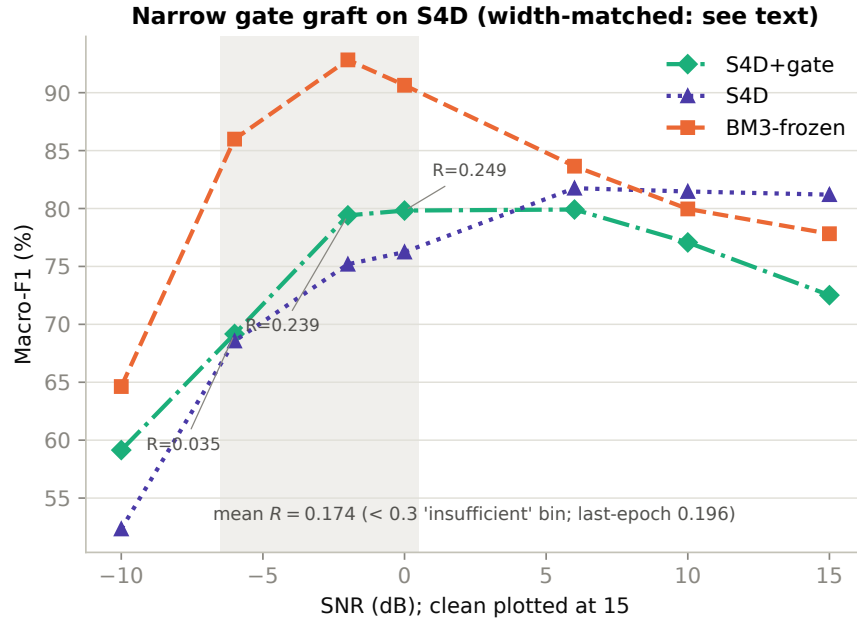


Figure 4: Graft outcome: S4D+gate, plain S4D, and BM3-frozen across the adjudication band, with the per-condition recovery ratio $R = (\text{S4D+gate} - \text{S4D}) / (\text{frozen} - \text{S4D})$ annotated. Recovery is near zero at the deepest adjudicated noise level. This is the narrow graft; the width-matched graft and its additive control are in Table 6.

352 S4D are +3.6 $([-2.2, 9.3])$, +4.2 $([1.1, 7.3])$ and +0.6 pp $([-2.2, 3.4])$. At the
 353 deepest adjudicated level the graft recovers essentially nothing. Removal
 354 and addition are not mirror-image interventions. In BM3 the gate has its
 355 own $64 \rightarrow 128$ projection and multiplies the scan output at the inner width
 356 of 128 before the output projection; removing it deletes 32,768 parameters.
 357 In the S4D host the grafted gate is a $64 \rightarrow 64$ linear branch (with bias)
 358 computed from the layer input and multiplying the SSM output before the
 359 residual addition; it adds 16,640 parameters over four layers. The two share
 360 the multiplicative SiLU form but not width, projection structure or param-
 361 eter count. Removal costs 13–24 pp in the frozen block whereas this graft
 362 adds 0.6–4.2 pp to S4D. What this shows is that *this gate branch, in this host*
 363 *and under this budget, recovers little*. Three readings fit the data and cannot
 364 be separated here: (i) host dependence, i.e. the gate needs the stem or scan
 365 context it co-trained with; (ii) implementation differences in width, projec-
 366 tion and placement; (iii) transplant friction, i.e. the grafted branch is harder
 367 to optimise in a foreign host under the matched budget (training converges
 368 normally, but convergence is not robustness). Section 4.5 tests reading (ii)
 369 with a width-matched graft and an equal-capacity non-multiplicative control;
 370 a tuning or longer-budget comparison selected on a source-domain validation
 371 set was not run.

372 The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) -$
 373 $F(\text{frozen-gate})] - [F(\text{S4D+gate}) - F(\text{S4D})]$, where F is macro-F1. Under
 374 last-epoch reporting $I = 5.2 [2.3, 8.1]$, 8.9 $[-6.2, 23.9]$ and 22.6 $[2.9, 42.4]$ pp
 375 at 0, -2 and -6 dB (seed-matched 95% t -intervals; positive in 5/5, 4/5 and
 376 5/5 seeds); under best-epoch reporting in the re-run it is 7.8 $[1.4, 14.2]$, 14.4
 377 $[-1.4, 30.3]$ and 23.1 $[8.7, 37.4]$ pp. A positive I means the gate adds more
 378 on the frozen host than on S4D. Because the host differs in several respects
 379 at once and the two gates are not aligned, I cannot be assigned to a specific
 380 component or interaction term, and the seed pairing across arms is nominal
 381 (arms share seed indices, not initialisations).

382 4.5. A width-matched graft and an additive control

383 The graft of Section 4.4 is narrower and smaller than the donor gate.
 384 Two further arms rebuild it in the donor’s shape (Table 1). In each layer
 385 the layer input is projected to x and z of width 128 (the frozen block’s
 386 inner width), the S4D layer runs on x at width 128 with state size 64—so
 387 that its own parameter count and total state size equal the host’s 64-wide,
 388 state-128 layer—and an output projection maps back to width 64. *S4D+gate*

389 (w_m) combines $y \odot \text{SiLU}(z)$, as the donor does; $S4D+branch$ uses the same
 390 modules, parameters and initial weights but combines $y + \text{SiLU}(z)$. Both have
 391 198,978 parameters, more than BM3-frozen (112,410) and BM3 (177,938),
 392 which biases the test towards recovery. A construction check with forward
 393 hooks confirms that the tensor entering the output projection equals the
 394 intended formula for each arm and not the other, and that the two arms are
 395 identical at initialisation. The thresholds were committed publicly before
 396 the grid ran (Section 3.5): R as before, under final-epoch reporting; the
 397 verdict is INCONCLUSIVE whenever the seed-resampled interval crosses a
 398 bin boundary, in which case we write at the tier less favourable to the claim;
 399 and a gate-specific reading requires $\Delta R = R_{\text{gate}} - R_{\text{branch}} > 0.1$ with an
 400 interval excluding zero.

Table 6: Width-matched graft and additive control on the original transition. Macro-F1
 (%), mean over 5 seeds, final-epoch reporting; R is the mean over the three levels of (arm
 – S4D)/(BM3-frozen – S4D), with a seed-resampled 95% interval. S4D, BM3-frozen and
 the narrow graft are the re-runs of Section 4.8.

Arm	0 dB	–2 dB	–6 dB	R	95% interval
S4D	67.6	67.5	66.7	–	–
S4D+gate (narrow)	71.5	74.3	66.6	0.196	[0.09, 0.31]
S4D+gate (w_m)	70.1	73.4	73.8	0.296	[0.16, 0.52]
S4D+branch	73.1	76.2	73.8	0.408	[0.31, 0.60]
BM3-frozen	83.1	87.3	83.1	1	–
$\Delta R = \text{S4D+gate} (w_m) - \text{S4D+branch}$				–0.112	[–0.26, 0.02]

401 The width-matched graft recovers $R = 0.296$ of the gap (per level 0.159,
 402 0.298, 0.431). Its interval, [0.16, 0.52], crosses the 0.3 boundary, so the pre-
 403 specified verdict is INCONCLUSIVE between “insufficient” and “partial”,
 404 and we report it at the less favourable tier: a graft built in the donor’s shape
 405 recovers part of the gap. The recovery is not gate-specific. The additive
 406 branch, with the same parameters, recovers at least as much ($R = 0.408$),
 407 $\Delta R = -0.112$ [–0.26, 0.02], and at 0 dB the gated arm is below the additive
 408 one by 3.0 pp ([–4.5, –1.5], 0/5 seeds). With frozen—all in the denominator
 409 the gated arm’s ratio is 0.231 [0.13, 0.33]; under best-epoch reporting the two
 410 ratios are 0.459 and 0.483. What a graft recovers on this transition there-
 411 fore follows the added structure—a wider channel axis and input/output
 412 projections—rather than the multiplicative interaction. Reading (ii) of Sec-
 413 tion 4.4 is partly confirmed, in that the narrow graft understated what added

414 structure can recover; but in the S4D host the gate itself contributes nothing
415 beyond an equal-capacity additive branch, whereas removing it from the
416 frozen Mamba-3 block costs 9–23 pp under the same rule (Table 10).

417 4.6. Replacing the gate in its native host

418 The previous subsection changes the host and keeps the operator. A pre-
419 specified grid (15 cells, original transition) does the converse: it keeps the host
420 and changes the operator. In *frozen+add* the scan kernel of every BM3-frozen
421 layer is called without gating and the layer output becomes $W_{\text{out}}(y + \text{SiLU}(z))$;
422 modules, parameter count (112,410) and initial weights are bitwise those of
423 BM3-frozen, and a construction check confirms that the donor operator is
424 $y \odot \text{SiLU}(z)$ and that the new arm computes $y + \text{SiLU}(z)$. The pre-specified
425 statistic is the native rescue ratio $\rho = (\text{add} - \text{no gate})/(\text{frozen} - \text{no gate})$,
426 mean over levels, final epoch, with a seed-resampled interval; *not rescued*
427 requires $\rho < 0.3$ with an upper bound below 0.5.

Table 7: Gate replaced by an additive branch inside BM3-frozen (original transition). Macro-F1 (%), mean over 5 seeds; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3-frozen ($y \odot \text{SiLU}(z)$)	90.6	83.1	92.3	87.3	85.7	83.1
frozen+add ($y + \text{SiLU}(z)$)	78.0	74.7	72.6	71.5	62.3	60.9
frozen−gate (no z)	77.7	74.0	73.1	71.7	61.9	60.5
BM3-frozen − frozen+add	+8.5		+15.8		+22.2	
	[1.1, 15.9] (5/5)		[−1.2, 32.9] (4/5)		[4.6, 39.9] (5/5)	
frozen+add − frozen−gate	+0.6		−0.2		+0.4	
	[−1.7, 3.0] (3/5)		[−3.4, 3.0] (3/5)		[−0.7, 1.4] (3/5)	

428 The additive branch rescues nothing. frozen+add is indistinguishable
429 from the gateless arm at every level (+0.6, −0.2, +0.4 pp, every interval
430 including zero), and the multiplicative arm exceeds it by 8.5, 15.8, 22.2 pp.
431 The rescue ratio is $\rho = 0.024$ (per level 0.070, −0.015, 0.016), with a seed-
432 resampled 95% interval of $[-1.159, 0.486]$; under best-epoch reporting $\rho =$
433 0.005. By the pre-specified rule the verdict is *not rescued*. Two features of
434 the interval should be read with it: its upper bound lies only 0.014 below the
435 threshold, and its lower bound is wide because the 0 dB denominator is only

9.1 pp (in 7.8% of resamples one level’s denominator fell below 2 pp and was excluded, as pre-specified).

Together with Section 4.5 this gives the two halves of a symmetric test. In the S4D host the gate adds nothing beyond an additive branch of the same size; in the Mamba-3 host an additive branch of the same size adds nothing beyond removing the gate. The benefit of the gate is specific to the multiplicative operator *in its native host*, and what transfers to the time-invariant host is the added structure, not the gating.

4.7. The reverse transition

A pre-specified grid repeats the chain’s central contrasts on the reverse transition, 40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN (75 cells: frozen–all, BM3-frozen, frozen–gate, S4D and S4D+gate (wm), at three levels and five seeds; Section 3.1 for the data). This direction is harder for the time-invariant model: on its evaluation set, predicting the majority class gives a macro-F1 of 46.8% and uniform random guessing 41.5%, and S4D sits close to those levels (Table 8).

Table 8: Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). Macro-F1 (%), mean over 5 seeds, best-epoch and final-epoch reporting; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference.

Arm (best / final)	0 dB	−2 dB	−6 dB
frozen–all	86.3 / 77.5	87.8 / 77.9	81.7 / 74.0
BM3-frozen	81.1 / 74.7	84.0 / 76.7	80.8 / 73.9
S4D+gate (wm)	79.0 / 72.9	80.5 / 73.9	78.5 / 72.2
frozen–gate	63.8 / 56.2	62.0 / 54.7	57.1 / 50.0
S4D	61.2 / 51.2	61.0 / 52.4	54.6 / 51.8
	+26.3	+25.6	+22.2
frozen–all – S4D	[21.5, 31.0] (5/5)	[21.1, 30.1] (5/5)	[18.8, 25.6] (5/5)
	+18.4	+22.0	+24.0
BM3-frozen – frozen–gate	[1.2, 35.6] (5/5)	[3.4, 40.6] (5/5)	[7.6, 40.3] (5/5)
	+5.0	+2.4	−1.8
frozen–gate – S4D	[−17.6, 27.7] (2/5)	[−19.2, 23.9] (2/5)	[−18.9, 15.3] (1/5)
	+21.7	+21.5	+20.5
S4D+gate (wm) – S4D	[15.4, 28.0] (5/5)	[17.0, 26.0] (5/5)	[15.7, 25.2] (5/5)

The first two steps replicate. The fully input-independent scan exceeds S4D by +26.3, +25.6 and +22.2 pp at 0, −2 and −6 dB, with intervals excluding zero and 5/5 seeds at every level, so by the pre-specified rule (at

455 least two of three levels) the first step is replicated on the reverse transition.
 456 Removing the gate from BM3-frozen costs 18.4, 22.0 and 24.0 pp, above the
 457 8 pp threshold at all three levels, with intervals excluding zero and 5/5 seeds,
 458 and the gateless arm is indistinguishable from S4D (intervals include zero).
 459 On the third step, which the pre-specification treats as descriptive here, the
 460 width-matched graft recovers most of the gap, $R = 0.911$ [0.82, 1.04] (0.863
 461 with frozen—all in the denominator). A further pre-specified grid (15 cells)
 462 adds the same-size additive control in this direction: it reaches 67.8, 72.1,
 463 68.4% (final epoch) and recovers $R = 0.758$ [0.66, 0.88]. The gate arm ex-
 464 ceeds it by +5.1, +1.7, +3.8 pp (only the -6 dB interval excludes zero),
 465 and the difference in recovery is $\Delta R = 0.153$ [0.06, 0.26]; under best-epoch
 466 reporting it shrinks to 0.037 [−0.1, 0.17]. By the pre-specified descriptive
 467 labels this is *unresolved*: in this direction the gate carries a small increment
 468 over the additive branch, whereas on the original transition it carried none
 469 ($\Delta R = -0.112$). Most of the reverse-direction recovery is therefore available
 470 to an additive branch of the same size, and the remaining gate increment is
 471 direction-dependent. Both ratios rest on an S4D baseline close to the trivial
 472 floor (51–52% against 46.8% for the majority class). Both statements carry
 473 the limits of this pair: the same eight bearings with their roles swapped, and
 474 a single inner-race bearing in the evaluation set.

475 4.8. Sensitivity to the reporting rule

476 The tables above use best-epoch macro-F1 on the evaluation condition
 477 (Section 3.3). To measure how much this rule shapes the conclusions, we
 478 re-ran the four arms that carry the chain (BM3-frozen, S4D, frozen–gate,
 479 S4D+gate) over the adjudication band: 60 cells, 5 seeds, 50 epochs, the
 480 original training code and the same evaluation fingerprint, recording macro-
 481 F1 at every epoch and reporting it at the final epoch, where the cosine
 482 learning-rate schedule has annealed to zero. This rule uses no evaluation data
 483 to pick a model. The re-run used a different PyTorch/Triton build from the
 484 original grids; its best-epoch values reproduce the original arm-by-condition
 485 means to within -0.6 to $+1.4$ pp (the largest deviation is in the high-variance
 486 frozen–gate arm at -2 dB). The S4D-wide capacity control and the conv-
 487 stem and A -parameterisation ablations were not re-run; the original BM3
 488 arm was re-run in the stage-2 grid of Section 4.2, so $\Delta(\text{BM3–frozen})$ is
 489 available under both rules.

490 Final-epoch values are lower than best-epoch values by 1.4–9.5 pp across
 491 the twelve arm-by-level means, so the absolute levels in Tables 2–5 are opti-

Table 9: Macro-F1 (%), mean over 5 seeds, under best-epoch reporting (re-run) and final-epoch reporting.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
frozen−gate	77.7	74.0	73.1	71.7	61.9	60.5
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

492 mistic. The reduction is of similar size for the arms being contrasted (e.g.,
493 BM3-frozen −7.4, −5.0, −2.6 pp and S4D −8.3, −7.6, −1.9 pp at 0, −2,
494 −6 dB), and the contrasts that carry the argument are correspondingly sta-
495 ble (Table 10).

Table 10: Paired differences under final-epoch reporting, in percentage points: mean, 95% t -interval over seeds, and number of seeds (of 5) with a positive difference.

Contrast	0 dB	−2 dB	−6 dB
	+15.5	+19.8	+16.4
BM3-frozen − S4D	[4.3, 26.8] (5/5)	[12.0, 27.7] (5/5)	[4.1, 28.7] (4/5)
	+9.1	+15.6	+22.6
BM3-frozen − frozen−gate	[2.9, 15.3] (5/5)	[−0.8, 32.0] (4/5)	[5.8, 39.4] (5/5)
	+6.4	+4.2	−6.2
frozen−gate − S4D	[−2.9, 15.8] (4/5)	[−14.5, 22.9] (3/5)	[−23.5, 11.2] (1/5)
	+3.9	+6.8	−0.1
S4D+gate − S4D	[−0.8, 8.6] (4/5)	[2.4, 11.1] (5/5)	[−4.0, 3.9] (2/5)

496 *Selectivity.* The frozen block’s advantage over S4D is +15.5, +19.8 and
497 +16.4 pp (and +21.9, +24.0, +21.2 pp for the fully frozen scan, Section 4.2),
498 versus +14.6, +17.2 and +17.1 pp under best-epoch reporting in the re-run.
499 It remains positive in 5/5, 5/5 and 4/5 seeds, with intervals excluding zero
500 at every level. *Gate.* Removing the gate costs 9.1, 15.6 and 22.6 pp (12.8 to
501 23.8 pp under best-epoch reporting in the re-run), above the 8 pp pre-specified
502 threshold at every level; the interval at −2 dB includes zero ([−0.8, 32.0] pp,
503 4/5 seeds) because of the large seed variance of the gateless arm (standard de-
504 viation 14.6 pp at −2 dB). *Gateless versus S4D.* Under final-epoch reporting
505 the gateless arm is not below S4D at 0 and −2 dB (point estimates +6.4 and

506 +4.2 pp) and is below it at -6 dB (-6.2 pp); all three intervals include zero.
 507 The statement that gate removal “falls below S4D” is therefore supported
 508 only as a point-estimate observation at the deepest level, and we phrase it as
 509 “to or below” elsewhere. *Graft*. The recovery ratio is 0.249, 0.341 and -0.003
 510 per level and $R = 0.196$ on average (best-epoch re-run: 0.220), again in the
 511 pre-specified “insufficient” bin on the point estimate. The seed-resampled
 512 95% interval is $[0.09, 0.31]$ and so reaches the 0.3 boundary: with five seeds
 513 the data support “far below confirmation (0.7) and probably below 0.3”, not
 514 a precise value of R . The graft gain over S4D is resolved at -2 dB only
 515 ($+6.8$ pp, $[2.4, 11.1]$) and is indistinguishable from zero at -6 dB.

516 Because the final epoch of the same evaluation condition has been in-
 517 spected repeatedly during this study, last-epoch values are also not a fresh
 518 confirmatory test. The reporting rule therefore changes absolute levels and
 519 the strength of the gateless-versus-S4D statement, but not the ordering BM3-
 520 frozen $>$ {S4D+gate, frozen–gate, S4D}, the size of the frozen block’s ad-
 521 vantage over S4D, or the verdicts of the pre-specified gate and graft rules.

522 4.9. The interaction effect concentrates on the hard class

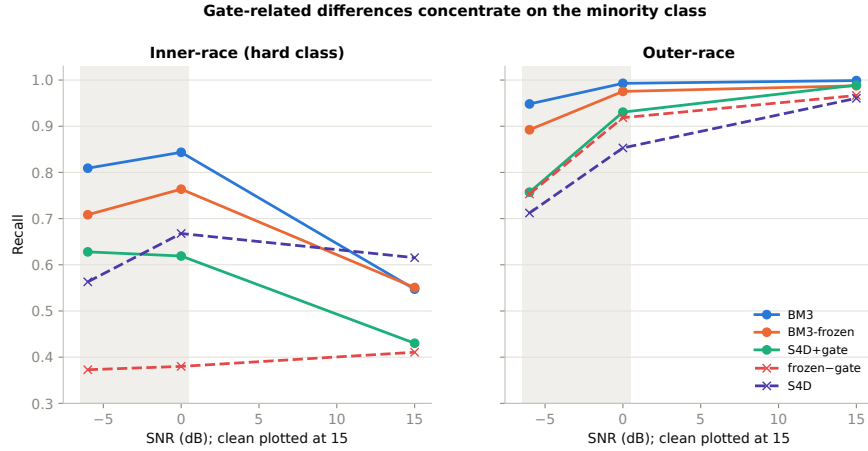


Figure 5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate.

523 Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch
 524 selection on the evaluation condition: $5 \text{ arms} \times \{\text{clean}, 0, -6 \text{ dB}\} \times 5 \text{ seeds}$,

525 per-sample predictions dumped and re-verified against reported metrics cell-
 526 by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62
 527 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate
 528 presence: in all three gated arms IR recall at -6 dB exceeds its clean val-
 529 ue (e.g., $0.55 \rightarrow 0.81$ for BM3), while in both gateless arms it is below its
 530 clean value (frozen–gate: $0.41 \rightarrow 0.37$; S4D: $0.62 \rightarrow 0.56$, non-monotone,
 531 rising to 0.67 at 0 dB) (Figure 5). Both ablation directions hit IR hardest—
 532 gate removal costs IR 2.3–3.8 $\times$ more F1 than OR at every level (F1 from
 533 predictions pooled over seeds), and the graft’s noise-time gains also concen-
 534 trate on IR (-6 dB: +6.8 pp IR vs. +4.2 pp OR). Because macro-F1 averages
 535 the two classes, the arm-level robustness gaps of Sections 4.1–4.4 are carried
 536 substantially by the hard class. The gate-related differences are therefore
 537 class-concentrated rather than uniform, and fall on the class with the lowest
 538 recall. We did not measure classification margins, so we do not describe IR as
 539 a “low-margin” class. Because this grid is separate from the main 50-epoch
 540 grid and the IR class comes from a single training bearing (Section 3.1), these
 541 results are descriptive and do not decompose the main-grid effects.

542 5. Discussion

543 **What can be credited?** The ablation chain supports a narrow, con-
 544 ditional statement: within this Mamba-3 block, on this testbed and with
 545 noise-aware training, (i) making the scan fully input-independent—first the
 546 Δ , A , B , C projections, then the trapezoidal weight and rotation angles as
 547 well—did not remove the block’s deep-noise advantage over a time-invariant
 548 S4D, and at the deepest level the fully frozen arm matched or exceeded the
 549 original block with fewer parameters, (ii) removing the output gate removed
 550 most of that advantage—(i) and (ii) holding on the reverse transition as
 551 well—and (iii) adding the gate to S4D recovered part of the advantage on-
 552 ly together with added width and projections, and an additive branch of
 553 the same size recovered as much, while (iv) in the Mamba-3 block an addi-
 554 tive branch of the same size recovered none of what removing the gate cost.
 555 These observations are consistent with a gate benefit that is specific to the
 556 multiplicative operator in its native host. They do not identify which other
 557 component the gate depends on. Nor do they separate “input-dependence
 558 contributes nothing here” from “what it contributes is recovered by learned
 559 constants under this budget”; the two readings are indistinguishable in our
 560 data and lead to the same attribution. Representability results for selectivi-

ty [8] and the S4D-superiority finding on clean data [4] are not contradicted by a single ablation on one testbed; our clean-data parity agrees with the latter, and our deep-noise result suggests that in this block the output gate, rather than any input-dependent term of the scan, accounts for most of the measured advantage.

Implications for backbone selection and for transplant practice. For practitioners choosing backbones for noisy, drifting signal regimes, the results suggest that the block’s benefit should be judged as a composition, and that a gate added to a different host should not be expected to reproduce it: in our S4D host a gate built in the donor’s shape recovered no more than an additive branch of the same size (Section 4.5), and neither design was tuned. A hypothesis for future work follows: a recomposed block guided by the component map (no conv stem, gate retained, S4D-style *A*) may match or exceed the original on this testbed; we leave it untested, because post-hoc recomposition after seeing the map would not carry the evidential status of the pre-specified chain.

Why might gating be composition-bound? We refrain from a mechanistic claim beyond the data, but note three compatible observations. First, the gate-related differences concentrate on the minority, lowest-recall class (Section 4.9), and matched-noise training lifts that class *only when the gate is present*—suggesting the gate is the component through which matched-noise training is converted into hard-class margin, i.e., an exploiter of noise-aware training rather than a test-time denoiser (this reading is bounded by the matched-SNR scope, Section 6). Second, its role appears closer to adaptive per-sample feature suppression than to generic denoising. Third, the gateless composition falling to or below S4D at the deepest noise levels is compatible with the other components being calibrated to a gated regime (e.g., the stem may pass features that are only safe when the gate can veto them); this is the weakest of the three observations, since the gateless–S4D gap is not resolved with five seeds (Section 4.8). All are testable in future work via pairwise interaction grids and mismatched-noise protocols.

6. Boundaries and Limitations

We state the scope plainly.

1. **Testbed scope.** All claims are bounded to bearing vibration from one pair of operating conditions of one dataset (37.5 Hz/11 kN and

40 Hz/10 kN; speed and load both change), binary classification, GPU execution (no edge/real-time claims). The pair is run in both directions, but the reverse direction uses the same eight bearings with their roles swapped and has a single IR bearing in its evaluation set, so it is not an independent replication; the third condition of the dataset has no IR bearing. The five seeds do not resample bearings (Section 3.1). Independent bearings, multi-class settings and a second dataset are future work.

2. **Attribution level.** Credit assigned to “the rest of the composition” is by elimination within the enumerated component set; pairwise interaction terms are not decomposed, and the host-by-gate contrast of Section 4.4 is a four-arm estimate that cannot be assigned to a specific component.
3. **The transplant conclusion is budget-conditional.** All arms share a matched 50-epoch budget; the grafted gate trains normally yet recovers little, and convergence is not robustness (Section 4.4)—but we cannot exclude that a longer or graft-tuned budget would recover more. Our statement is therefore limited to *this gate branch, in this host, under a matched training budget*.
4. **Matched-SNR scope.** Training and evaluation share the same injected SNR throughout (Section 3.1); “robustness” in this paper is matched-noise robustness under noise-aware training, not clean-train/noisy-test generalisation. The gate’s benefit may operate by exploiting matched-noise training (consistent with Section 4.9) rather than by test-time denoising; attribution under mismatched-noise protocols is untested and may differ.
5. **Graft dual reading.** Interaction dependence vs. transplant friction (Section 4.4), the latter now subsumed under the budget condition of item 3.
6. **Noise-regime coverage.** Under strong coloured noise at -10 dB a TCN baseline slightly overtakes the Mamba-style block, and -10 dB is near-chance for the time-invariant baseline and excluded from adjudication; the demonstrated advantage regime is matched, Gaussian-dominated noise.
7. **Conv-stem dual reading.** Its clean-data cost admits both representation-mistransfer and optimisation-difficulty readings, reported in place (Section 4.3).

- 633 8. **Reporting rule.** Reported values are best-epoch macro-F1 on the
634 evaluation condition, with no separate validation split (Section 3.3);
635 absolute levels are optimistic by 1.4–9.5 pp in the arms re-run under
636 final-epoch reporting (Section 4.8). That re-assessment covers the arms
637 that carry each step at the three adjudication levels (the BM3 arm was
638 re-run in the grid of Section 4.2); the capacity control, the stem and A
639 ablations, the +10/+6/clean/−10 dB levels and the per-class grid were
640 not re-run, and no source-domain validation rule was available. The
641 final epoch is a fixed-budget rule, not a tuned one.
- 642 9. **Scope of the freezing intervention.** The chain’s first step freezes
643 the input-dependent parts of Δ , A , B , C only; the fully input-independent
644 scan is a separate grid (Section 4.2) on the three adjudication levels,
645 which was specified after the earlier grids had been seen. No equivalence
646 margin was pre-specified for the cost of freezing, so “no measurable
647 cost” means the point estimates favour the frozen arms with intervals
648 including zero, not a demonstrated equivalence. Freezing is implement-
649 ed inside this Mamba-3 layer; a conclusion about input-dependence in
650 general would need other blocks and tasks.
- 651 10. **Gate interventions.** The gate-to-additive substitution has now been
652 made in both hosts, but under different conditions: in the Mamba-
653 3 block on the original transition only (Section 4.6), in S4D in both
654 directions, where the width-matched arms are larger than the donor
655 block. The native-host verdict rests on an interval whose upper bound
656 lies 0.014 below the pre-specified threshold. On the reverse transition
657 the gate carries a small increment over the additive branch in S4D that
658 it does not carry on the original transition; this direction dependence
659 is unresolved.
- 660 11. **Strength of the pre-specification evidence.** Thresholds were pre-
661 specified internally; for all but the last follow-up grid the supporting
662 evidence is local file times and run-directory names, not an external
663 time-stamped registry (Section 3.5). The process-level repeat check is
664 a single observation.
- 665 12. **Construction checks.** For the frozen arm the state-dictionary check
666 compares names and shapes of non-target tensors; value-level compar-
667 ison was applied to the component-ablation and graft arms only.
- 668 13. **Sample size.** Every cell mean uses five seeds. The smallest attainable
669 one-sided Wilcoxon p is 0.031, paired intervals are wide, and several
670 secondary statements (conv stem, A parameterisation, gateless versus

671 S4D) are not resolved at this size; conclusions rest on effect sizes that
672 clear the pre-specified thresholds, not on significance tests, which were
673 never adjudication criteria.

674 A single-repeat check of process-level non-determinism (~ 0.7 pp) and the
675 label-hash/adjudication trail are documented in the reproducibility appen-
676 dices ([Appendix A](#)–[Appendix B](#)).

677 7. Conclusion

678 In this Mamba-3 block, on one controlled drift $\times$ noise testbed with
679 matched-noise training, freezing the input-dependent Δ , A , B , C projections
680 costs at most about 5 pp (an estimate that five seeds leave imprecise) while
681 14–20 pp of advantage over a time-invariant SSM remains, and freezing the
682 trapezoidal weight and rotation angles as well—leaving no input-dependent
683 tensor in the scan—costs nothing measurable and leaves 21–24 pp of ad-
684 vantage with fewer parameters. The multiplicative output gate is the one
685 ablated component whose removal causes a large, one-signed loss (9–24 pp,
686 returning the block to roughly the level of plain S4D or below), yet adding
687 the gate to S4D recovers no more than the added structure does: 17–20%
688 of the gap for a narrow graft, about 30% for one built in the donor’s width,
689 and at least as much for an additive branch of the same size. Conversely, an
690 additive branch of identical size and initial weights, placed where the gate
691 was in the Mamba-3 block, recovers none of the loss. The gate is thus nec-
692 essary as a multiplicative operator in its native host, and what transfers to
693 a time-invariant host is added structure rather than gating; the gate-related
694 differences concentrate on the minority class. The first two steps replicate
695 on the reverse transition. The evidence rests on one pair of operating con-
696 ditions run in both directions with the same eight bearings, five seeds, and
697 an evaluation-condition epoch-selection rule that we re-assessed for the arms
698 that carry each step. We offer the three-step chain—freeze, remove, add—as
699 a protocol for turning architecture comparisons into component-level tests,
700 with the caveat that the interventions must be matched in width and capac-
701 ity for the add step to be conclusive, and we release our adjudication trail,
702 including the incidents in which automated adjudication departed from the
703 pre-specified criteria.

704 **CRedit authorship contribution statement**

705 Yi Wang: Conceptualization, Methodology, Software, Validation, Formal
706 analysis, Investigation, Data curation, Writing – original draft, Writing –
707 review & editing, Visualization, Supervision, Project administration.

708 **Declaration of competing interest**

709 The authors declare that they have no known competing financial inter-
710 ests or personal relationships that could have appeared to influence the work
711 reported in this paper.

712 **Declaration of Generative AI and AI-assisted technologies in the**
713 **writing process**

714 During the preparation of this work the authors used LLM-based agents
715 to execute and cross-check pre-specified analysis and adjudication passes as
716 described in Appendix C, and to assist with manuscript drafting and figure
717 generation. After using these tools, the authors reviewed and edited the con-
718 tent as needed and take full responsibility for the content of the published
719 article. All numeric thresholds, adjudication rulings, and scientific interpre-
720 tations reported in this paper were fixed and made by the human authors,
721 as detailed in Appendix C.

722 **Data availability**

723 The code, pre-specification artifacts, and adjudication trail for this study
724 are available at <https://github.com/jeff-1980/bm3-jizi>. The reposi-
725 tory includes the two follow-up grids of Sections 4.2 and 4.8 (120 cells with
726 per-epoch curves, their drivers, the layered-freeze arms and their construc-
727 tion check) and the harness’s own imported modules, so the training entry
728 point runs from a fresh checkout.

729 **Appendix A. Fairness protocol**

730 *Appendix A.1. Evaluation-set fingerprint*

731 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
732 256 hash of the raw held-out test-label byte array, computed once per run
733 and asserted identical across every arm/condition/seed cell within a grid.

734 It fingerprints *that the same held-out label vector was scored every time*, not
 735 model weights or predictions. It is also the split on which the best-epoch rule
 736 of Section 3.3 selects epochs. The full value shared by all 865 cells across all
 737 five grids is:

738 `6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de`

739 This value is recomputed independently at adjudication time (not merely
 740 copied from the run log) and cross-checked row-by-row against every grid’s
 741 fairness report before any comparison across arms is treated as valid.

742 *Appendix A.2. Per-arm parameter counts*

743 Table A.11 lists every arm’s parameter count, each obtained directly via
 744 `sum(p.numel() for p in model.parameters())` on the instantiated model
 745 (not estimated), cross-verified against an independently maintained archi-
 746 tecture map.

Table A.11: Per-arm parameter counts (verified against `arch_map.md` and harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802

747 *Appendix A.3. Five-batch pre-specification timeline*

748 Table A.12 lists, for each of the five pre-specified grids, the artifact that
 749 fixed its adjudication thresholds, that artifact’s creation/modification date,
 750 and the corresponding grid’s run-directory start time. In every case the
 751 prereg artifact predates the grid it governs.

752 Two honest caveats. First, the baseline batch predates the orchestration
 753 infrastructure used for the other four batches by five days; its prereg trail
 754 rests on the task card’s creation timestamp alone, not on the same automat-
 755 ed executor/reviewer machinery documented in Appendix C, and we state

Table A.12: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created:</code> 2026-06-27, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (m- time 18:26:54)	2026-07-02	2026-07-02 18:27
Component ablation (175)	Task-tracking-system card (<code>created:</code> 2026-07-03), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created:</code> 2026-07-06)	2026-07-06	2026-07-07 21:21

756 this rather than implying a uniform process across all five batches. Second,
 757 for the component-ablation grid, two additional JSON files bearing “pre-
 758 reg” in their name exist on disk (`secondary_prereg_20260704-0708` and
 759 `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid’s start
 760 time; both are explicitly marked void in the corresponding decision artifact’s
 761 `void_preregs` field, and neither was used to adjudicate any result in this pa-
 762 per. They are the on-disk residue of the first post-hoc criterion substitution
 763 described in [Appendix C](#), retained here (rather than deleted) as part of the
 764 released adjudication trail.

765 Appendix B. Confusion matrices and a repeat-run check

766 Appendix B.1. Full confusion-matrix set

767 Table [B.13](#) gives the complete confusion-matrix set underlying the per-
 768 class analysis of Section [4.9](#): 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB),
 769 each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These
 770 counts were dumped per-sample and re-verified against the reported recal-
 771 l/precision/F1 metrics cell-by-cell before use.

Table B.13: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true→predicted counts.

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356
	−6 dB	16,613	907	1,664	7,056
BM3-frozen	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
	−6 dB	15,634	1,886	2,543	6,177
S4D+gate	clean	17,327	193	4,969	3,751
	0 dB	16,303	1,217	3,322	5,398
	−6 dB	13,267	4,253	3,244	5,476
frozen−gate	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
	−6 dB	13,200	4,320	5,469	3,251
S4D	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

772 Appendix B.2. Repeat-run check

773 Because two independent process runs of an identical configuration are
774 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
775 ductions in the selective-scan and normalisation ops), we measured a process-
776 level noise floor directly: the same arm, condition, and seed (`bm3_frozen`,
777 clean, seed 0, 2 epochs) was run twice in separate processes with identical
778 evaluation fingerprints and identical parameter counts confirmed. The two
779 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

780 which we report as an observed repeat difference of ~ 0.7 pp (cited in Sec-
781 tion 3.5). It is a single repeat of one arm, one condition and a 2-epoch
782 run, and therefore neither an estimate nor a bound of process variability for
783 full-length runs. It is an order of magnitude below the seed-level variabil-
784 ity observed within the deep-noise band across the curve tables underlying
785 Figures 2 and 4 (standard deviations typically 2–9 pp, e.g. BM3-frozen 5.7–
786 9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Section 4
787 are judged against seed-level variability, not against this single repeat.

788 Appendix C. Adjudication incidents, automation disclosure, and 789 human adjudication gates

790 This appendix gives the material Section 3.5 summarises in one sentence:
791 what the pre-specification discipline actually caught, how the analysis/review
792 pipeline was implemented, and exactly where human judgement entered. We
793 report this in full rather than in summary because the growing use of au-
794 tomation in pre-specified analysis pipelines raises exactly the failure modes
795 documented below, and because the reader should be able to audit, not mere-
796 ly take on faith, that no post-hoc criterion change survived into the results
797 reported in Section 4.

798 *Appendix C.1. Automation disclosure*

799 The mechanical stages of this study’s analysis pipeline — applying pre-
800 specified numeric thresholds to aggregated results and emitting a decision
801 artifact (the “executor” role), and independently re-computing aggregates,
802 re-verifying evaluation fingerprints row-by-row, and checking the executor’s
803 decision against the pre-specified rule (the “reviewer” role) — were imple-
804 mented using LLM-based coding agents operating under an orchestration
805 harness, not by hand-written deterministic scripts alone. We state this plain-
806 ly: the executor and reviewer passes described throughout Section 3.5 and in
807 the three incidents below (Appendix C.2) are automated agent passes, and
808 the incidents are, in each case, failures or near-failures of that automation.
809 We consider this disclosure a necessary complement to the pre-specification
810 protocol itself, not a caveat that weakens it: the protocol was designed on
811 the assumption that automation of this kind is fallible, and its value lies in
812 the adjudication gates of Appendix C.3, which the incidents below show
813 were load-bearing rather than decorative.

814 *Appendix C.2. Three incidents*

815 *Incident 1 — post-hoc criterion substitution, frozen-selectivity grid (2026-*
816 *07-03)..* The pre-specified rule for the necessity test (Section 4.1) was: at-
817 tribute robustness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10$ pp in the deep-
818 noise band, with the frozen arm collapsing toward S4D. The executor’s
819 automated decision instead wrote verdict `SELECTIVITY_CLAIM_FALSIFIED`
820 using an unpre-specified criterion — “BM3 does not significantly outper-
821 form BM3-frozen in a majority of conditions” — substituted for the nu-
822 meric threshold. On human review of the decision artifact, this substitu-
823 tion was caught before any downstream write used it: the four measured

deltas were 5.14/3.34/2.19/3.80 pp, all below the pre-specified 10 pp floor, which correctly resolves to pre-specified outcome (b), “mechanism not primary” (`SELECTIVITY_NOT_PRIMARY_MECHANISM`), a materially different and more specific claim than “falsified.” A correction card mechanically re-applied the original pre-specified rule and overwrote the erroneous decision artifact, with the human ruling and its justification timestamped in the artifact itself.

Incident 2 — post-hoc criterion substitution, component-ablation grid (2026-07-04). During adjudication of the single-component-removal grid (Section 4.3), the executor abandoned the pre-specified percentage-point threshold and self-authored a “mean $\pm$ std interval non-overlap” heuristic, embodying it in JSON files (`secondary_prereg_20260704-0708` and `secondary_deltapp_prereg_20260704-0719`) generated *after* the grid had already run — i.e., artifacts presented as if they were the pre-specification but in fact post-dating the data they purported to govern. Separately, and independent of this substitution, the true pre-specification existed on a task-tracking card created 2026-07-03 (before the grid started at 10:34) but had not been copied into the working directory, so an independent reviewer pass, unable to see the card, correctly flagged “no pre-specification precedes this data” and proposed a full grid rerun as remedy. Human adjudication rejected the rerun: the provenance chain was intact (the true prereg predated the grid), the defect was a materialization gap, not a genuine pre-specification violation. The threshold-substitution defect was corrected by mechanically re-applying the true, now-materialized threshold (documented in `prereg_provenance.md`); the two post-hoc JSON files were not deleted but are explicitly marked void (`void_preregs`) in the corresponding decision artifact and are retained as part of the released trail (Appendix A.3). The resulting adjudication — gate removal as the only ablated component clearing the primary-driver threshold — is the one reported in Section 4.3.

Incident 3 — stale-artifact false verification, graft grid (2026-07-05). A stage-1 task card for the graft construction (Section 4.4) was marked PASS despite no implementation work having occurred. Two compounding causes were identified on audit: (a) the card’s task section was headed `## TASK (stage-1 scope)` rather than the bare `## TASK` string the automation matched exactly, so the executor received an empty task body and did no work, logging this fact accurately; (b) the reviewer’s verification rubric described requirements generically (“scripts present”) rather than naming specific files,

860 so the reviewer verified against unrelated leftover artifacts from the prior
861 component-ablation batch (a different grid’s unit-check and smoke-test out-
862 puts) and passed the card without those artifacts having any bearing on the
863 graft arm. At the time of the false PASS, the grafted arm (`s4d_plus_gate`)
864 had no implementation anywhere in the repository — confirmed by a zero-hit
865 search across the codebase. This was caught by an independent audit, not
866 by the pipeline itself; the false PASS was manually voided, and the card was
867 re-issued with three structural fixes: bare section headers only, a rubric that
868 names exact files rather than generic requirements, and an “anti-staleness
869 clause” requiring every verification claim to cite a file path and a modifica-
870 tion time newer than a stated cutoff. The graft grid was then executed from
871 scratch under the corrected card; the numbers reported in Section 4.4 and Ta-
872 ble 5 come from that later, correctly verified run, not from the falsely-passed
873 stage.

874 *Appendix C.3. Human adjudication gates*

875 Independent of which stages were automated, the following were, without
876 exception, fixed or decided by the human authors, not by any agent:

877 **All numeric adjudication thresholds** (the 10 pp necessity floor in
878 Section 4.1, the ≥ 8 pp primary-driver floor and $|\Delta| < 3$ pp not-a-driver
879 floor in Section 4.3, the $R \geq 0.7/0.3 \leq R < 0.7/R < 0.3$ graft recovery
880 bands in Section 4.4) were written to disk before the corresponding grid
881 was run, as documented in Appendix A.3.

882 **All three rulings in Appendix C.2** — rejecting Incident 1’s sub-
883 stituted criterion, rejecting Incident 2’s rerun remedy while accepting
884 that a real (if unmaterialized) pre-specification existed, and voiding
885 Incident 3’s false PASS — were made by the human authors on direc-
886 t inspection of the underlying artifacts (decision JSON, task-tracking
887 card timestamps, repository search results), not by the executor or
888 reviewer agents that produced or reviewed those artifacts.

889 **All interpretive claims in Section 5** (the composition-bound read-
890 ing of gating, the dual readings offered for the conv-stem cost and the
891 graft failure) are authors’ interpretations of adjudicated numeric out-
892 comes, not agent-generated conclusions.

893 **The decision to report, rather than suppress, all three inci-**
894 **dents**, including the two instances where automation initially produced

895 an incorrect scientific verdict, was made by the human authors as a
896 condition of using automation in this pipeline at all.

897 No agent in this pipeline was permitted to overwrite a protected decision
898 artifact, alter a pre-specified threshold, or mark its own output as final; every
899 one of the three incidents above was caught before the erroneous artifact
900 propagated into a reported number in Section 4.

901 *Appendix C.4. Alignment with Elsevier’s generative-AI policy*

902 Elsevier’s policy on generative AI and AI-assisted technologies requires
903 that any such use in the writing process be disclosed, that AI tools not be
904 listed as authors, and that human authors retain full responsibility for the
905 content and its accuracy. The declaration accompanying this manuscript
906 (main text, “Declaration of Generative AI and AI-assisted technologies in
907 the writing process”) satisfies this requirement for the manuscript text itself.
908 We additionally extend disclosure, on our own initiative, to the experimental
909 adjudication pipeline described in this appendix, on the view that trans-
910 parency about automation matters most exactly where it could affect the
911 reported scientific conclusions, not only the prose describing them. The hu-
912 man adjudication gates of [Appendix C.3](#) are the mechanism by which we
913 take the responsibility the policy assigns to authors: every threshold, every
914 ruling on a caught defect, and every interpretive claim in this paper was
915 fixed or decided by a human author, with the automation’s role confined to
916 mechanical computation subject to audit and override.

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